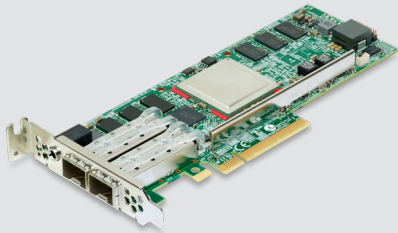


Product Brief



Key Features

- PCIe NIC in half-height, half-length form factor
- PCIe 3.0 x8 Host Interface
- Stingray data center SoC: eight 64-bit ARM Cortex-A72 cores at 3.0 GHz with 16 MB cache
- Integrated NetXtreme 100GbE NIC
- TruFlow configurable flow accelerator
- RoCE v1/v2 support
- SR-IOV support
- 4, 8, or 16 GB of onboard dual-channel DDR4 memory
- 100 Gb/s cryptography engine with single-pass hashing and encryption/decryption
- 100 Gb/s RAID5, RAID6, and erasure engine
- Deduplication (De-Dup) including Rabin Fingerprint, SHA, and MD5
- Secure boot
- Secure key storage
- ARM TrustZone
- Public Key Acceleration (PKA) engine

Stingray™ PS225

2×25-Gb High-Performance Data Center SmartNIC

Overview

The Broadcom® Stingray™ PS225 SmartNIC is purpose-built to enable solutions in the evolving data center. The board integrates powerful networking and processing subsystems for SmartNIC applications including dataplane acceleration and software-defined storage (SDS) including NVMe-over-Fabrics (NVMe-oF). The PS225 is designed in a low-profile, half-length form factor for both half-height and full-height server applications.

The PS225 SmartNIC features the Stingray high-performance data center SoC, which incorporates a 100G NetXtreme® E-Series advanced network controller, a high-performance ARM CPU block, PCI Express (PCIe) 3.0 x8 interface, key accelerators for compute offload, and a high-speed memory subsystem including L3 cache and DDR4 interfaces—all interconnected by a coherent Network-on-Chip (NoC) fabric.

The network interface is powered by the latest generation of NetXtreme controller technology, delivering a high packet rate and low latency. Support for the NVMe-oF standard is enabled over RoCE (RDMA over Converged Ethernet) v1 and v2 as well as over TCP and other transports, ensuring a future-proof and scalable platform for the deployment of disaggregated storage over a range of topologies. The powerful TruFlow™ configurable flow accelerator adds a powerful packet inspection and processing capability in the hardware, moving common flow-processing workloads into hardware and freeing CPUs for application workloads. The ARM CPU subsystem features eight ARMv8 Cortex-A72 CPUs at 3.0 GHz, arranged in a multi-cluster configuration.

The PS225 Ethernet interface uses a multirate SerDes PHY that is common to Broadcom switch silicon connected to two SFP28 connectors to provide dual-port 25-Gb connectivity. The interface supports direct connection of optical modules or DAC (direct-attach copper) cables.

The Stingray family of products is supported by a comprehensive Software Development Kit providing a complete turn-key Linux distribution, and the open programming model enables application porting and development with minimal effort. The platform uses standard Linux and standard GNU tools and libraries. All drivers for the Stingray SoC are upstreamed to kernel.org and all peripheral interfaces use standard device drivers.

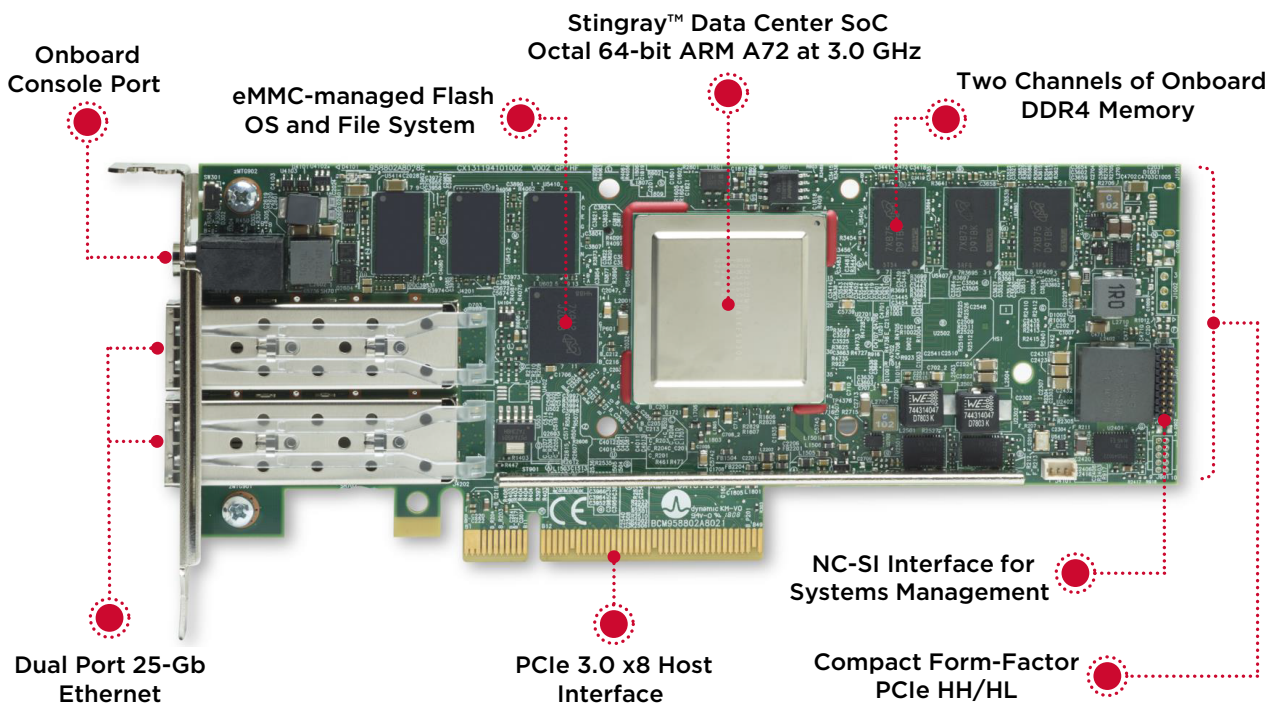
Applications

- Smart NIC
- Bare Metal Services
- Storage Services
- Network Function Virtualization (NFV)

Benefits

- The industry's first 2x25G SmartNIC with high-performance 3 GHz ARMv8 processors featuring an integrated NetXtreme Ethernet controller with TruFlow packet processing engine, congestion control, multi-host capability, and RoCEv2.
- The industry's lowest power data center SmartNIC, enabling integration into industry standard server platforms including PCIe half-height, half-length form factor.
- Advanced SmartNIC adapter with highly efficient hardware acceleration engines.
- Integrates encryption/decryption offload engines with a 100G full-featured NetXtreme NIC with TruFlow to accelerate and offload intensive packet processing and application workloads from the host CPU to increase performance and reclaim CPU cycles for host applications.
- Incorporates RoCEv2 and SR-IOV enhancing network data transfers and I/O performance.
- Compact, low power, high-performance 50G data center SmartNIC.
- Optimal power efficiency and performance enabled by a 16-nm FinFET+ process.
- Industry-leading octal-core 3 GHz ARMv8 Cortex-A72 CPU architecture.

Stingray PS225 Board



Ordering Information

BCM958802A8044C	PS225-H04 dual-port 25 Gb/s Smart NIC adapter with 4 GB DDR4 memory and 16 GB eMMC
BCM958802A8048C	PS225-H08 dual-port 25 Gb/s Smart NIC adapter with 8 GB DDR4 memory and 16 GB eMMC
BCM958802A8046C	PS225-H16 dual-port 25 Gb/s Smart NIC adapter with 16 GB DDR4 memory and 64 GB eMMC